

Tracking Topic Evolution Over
Time - Temporal Analysis of
COVID-19 Research Trends

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Introduction

A major challenge in COVID-19 research is the dynamic and constantly changing nature of its focus. As new data emerges and global events unfold, scientists must quickly adjust their research priorities. This constant evolution makes it difficult to track which research areas are prioritized and how major events influence scientific research. Our study addresses this gap by analyzing the progression of COVID-19 research themes over time. By mapping these trends, we reveal how the scientific community responds to real-world challenges by uncovering patterns that might otherwise go unnoticed. These insights do more than advance academic understanding, they provide actionable insight for policymakers and researchers, which allows us to identify key shifts and ensures that resources and efforts are directed where they are needed most.

In our project, we used Latent Dirichlet Allocation (LDA) for topic modeling, which enabled us to extract and analyze key research topics over time. This method allowed us to track the evolution of research themes, which provided us with valuable insights into how the focus of COVID-19 research has changed in response to emerging events and data.

Analyzing the evolution of COVID-19 research topics allows us to get critical insights that significantly inform research priorities, policy adaptation, and resource allocation. Identifying prevailing trends and gaps in scientific research helps guide research priorities and enables academic and funding institutions to focus efforts on high-impact and underexplored areas, ultimately improving research efficiency. Moreover, the ability to monitor shifting research themes enables policymakers and healthcare leaders to adapt strategies dynamically, and it ensures that public health measures are both responsive and grounded in the latest evidence. Additionally, a deep understanding of these evolving trends promote strategic resource distribution, which guides the allocation of funding, personnel, and materials to areas of most urgent need. Ultimately, this research not only improves the effectiveness of the current pandemic response but also strengthens preparedness for future health crises ensuring that responses are more informed and targeted.

Methodology

The successful temporal analysis of COVID-19 research trends using LDA depends on a strong data preprocessing framework. This phase is designed to clean and structure the dataset to capture the evolution of scientific research throughout the pandemic. We began with using the metadata.csv from the COVID-19 collection. During the initial data assessment in Pandas framework, we focused on identifying key columns, such as title, abstract, publish_time, and authors. These keys provided us with the textual data for topic modeling and temporal reference points that we used for trend analysis. However, there were records lacking an abstract or publish_time that were excluded by dropna() to ensure that further analysis could be performed. Handling the publish_time column with pd.to_datetime, we could manage the removal of entries

with unparseable data formats, which strengthened the temporal aspect. Later, the dataset was modified to only include publications from 2020 to 2025, as these captured the height of research activity during the pandemic. Despite this, the dataset version used acknowledges that data sparsity extends beyond this range. Furthermore, the `year_quarter` feature developed with `dt.to_period('Q')` provided the necessary level of detail to track shifts in themes according to the quarter.

Due to the large computational cost associated with large-scale embedding and clustering, coupled with the possibility of temporal bias caused by fluctuating publication columns, a strategic sampling approach was required. To minimize computational requirements, stratified random sampling was implemented in place of processing the entire relevant data (limiting the analysis to approximately 50,000 records). This helped mitigate bias while remaining within the computational constraints.

Following this, the textual data within this subset was cleaned and normalized using NLTK tools after being concatenated into the `processed_text` column to maximize context. As a part of the framework, all text was converted to lowercase, digits were removed with regular expressions (`re.sub`), punctuation was effectively eliminated by using `RegexpTokenizer(r' /w+')`, common English stopwords were filtered out using `nltk.corpus.stopwords`, and `WordNetLemmatizer` was used to lemmatize the text because lemmatization reduces words to their dictionary form which preserves semantics more effectively. The models used for downstream semantic embedding rely heavily on this factor. Additionally, we removed very short tokens as they were likely to be noise. By clearing the textual data comprehensively, the signal-to-noise ratio was maximized.

To ensure that the language models had access to the most contextually complete input, a preprocessing function was applied separately to both the title and abstract columns. These were stored in `processed_title` and `processed_abstract` respectively. Afterwards, the two cleaned strings were concatenated into a new column `processed_text`. This allowed the model to benefit from both the concise thematic signal of the title and the expanded description within the abstract. This concatenated text served as the final input for embedding, which ensured richer semantic representation during topic modeling.

To analyze how COVID-19 research evolved throughout the pandemic, we developed a systematic approach that combined temporal analysis with natural language processing. First, we organized the articles by their publication dates, grouping them into quarterly segments. This chronological framework was essential to track the shifts in research priorities, from the initial outbreak to developments like vaccine deployment and variant emergence, reflecting the rapid changes in the pandemic's evolution.

For text processing, we used TF-IDF vectorization to transform the article titles and abstracts into analyzable numerical data. We adjusted the TF-IDF parameters, specifically

max_df to exclude terms appearing in nearly all documents and min_df to exclude terms appearing in only a few, which ensured the focus remained on statistically significant content.

The core of our analysis involved using Latent Dirichlet Allocation (LDA) for topic modeling to track the evolution of COVID-19 research themes over time. Recognizing that a static model couldn't fully capture the dynamic nature of the research, we customized our approach for each quarterly segment. We tested various numbers of topics in each period and selected the optimal model through an iterative process based on coherence scores. These scores, which measure the semantic similarity between the most relevant words within a topic, served as indicators of the topic's clarity and usefulness. By carefully determining the number of topics based on the maximum coherence scores for each quarter, we ensured that each model produced clear and meaningful topics that accurately reflected the evolving nature of COVID-19 research.

```
Quarter: 2020Q1, Optimal Number of Topics: 10, Highest Coherence Score: 0.4182
Quarter: 2020Q2, Optimal Number of Topics: 6, Highest Coherence Score: 0.5079
Quarter: 2020Q3, Optimal Number of Topics: 10, Highest Coherence Score: 0.5167
Quarter: 2020Q4, Optimal Number of Topics: 9, Highest Coherence Score: 0.5248
Quarter: 2021Q1, Optimal Number of Topics: 5, Highest Coherence Score: 0.5412
Quarter: 2021Q2, Optimal Number of Topics: 6, Highest Coherence Score: 0.5194
Quarter: 2021Q3, Optimal Number of Topics: 8, Highest Coherence Score: 0.4611
Quarter: 2021Q4, Optimal Number of Topics: 10, Highest Coherence Score: 0.5245
Quarter: 2022Q1, Optimal Number of Topics: 8, Highest Coherence Score: 0.5338
Quarter: 2022Q2, Optimal Number of Topics: 10, Highest Coherence Score: 0.4500
Quarter: 2022Q3, Optimal Number of Topics: 6, Highest Coherence Score: 0.5058
Quarter: 2022Q4, Optimal Number of Topics: 10, Highest Coherence Score: 0.4855
```

To refine our model, we computed the average of the most frequently optimal topic numbers derived from our prior tests as you can see from the image above. This calculation revealed that the optimal number of topics across quarters was 8. Using this optimal number of topics, we retrained the LDA model on a TF-IDF matrix. This process captured the significance of key terms while filtering out common or irrelevant words that do not contribute meaningful insights.

The LDA model was trained using online learning, which allowed efficient processing of our large dataset. This was crucial for scalability and faster processing, since we were working with thousands of COVID-19 research articles. After training, we identified the most important words for each topic by sorting them based on their significance and displayed the top 10 words for each topic, which you can see from the image below.

Topic 1: • student • learning • education • teaching • hcws • cancer • medical • online • virtual • opioid	Topic 3: • care • patient • pandemic • covid • surgery • healthcare • service • surgical • hospital • nurse	Topic 5: • vaccine • vaccination • systematic • systematic review • review • covid vaccine • meta • meta analysis • covid vaccination • mrna	Topic 7: • learning • image • model • pregnant • machine • pregnancy • deep • machine learning • woman • algorithm
Topic 2: • covid • health • pandemic • data • model • social • research • country • study • public	Topic 4: • patient • covid • case • disease • infection • mortality • risk • clinical • day • severe	Topic 6: • mental • anxiety • mental health • health • child • participant • depression • study • stress • adult	Topic 8: • cov • sars • sars cov • cell • protein • virus • infection • viral • antibody • drug

This allowed us to interpret the key themes present in the research. By analyzing the topics generated by LDA, we gained valuable insights into the topic evolution of COVID-19 research, which reveals how the scientific community adapted to new challenges and global events over time.

To dive deeper into the semantic content of the research articles, we used transformer-based embeddings for advanced textual analysis. The transformer library allows the use of SciBERT, which contains a distinct advantage over generic models in understanding vocabulary and complex relationships within research texts. By processing the processed_text in batches and using GPU acceleration with Torch when available, we achieved more efficient embeddings.

Clustering algorithms face challenges because of the high dimensionality of SciBERT embeddings (“the curse of dimension”). This problem was addressed using Uniform Manifold Approximation and Projection (UMAP) from the UMAP learning library. In comparison with alternatives, UMAP has been identified as a strong performer for preserving both local and global data structures. With this method, embedding was reduced to 10 dimensions while still maintaining meaningful topological relationships. Using these parameters, we were able to create an optimized input for the subsequent clustering step with random_state=42 to ensure reproducibility. It balances the representation of local neighbourhood structures and broader cluster separation.

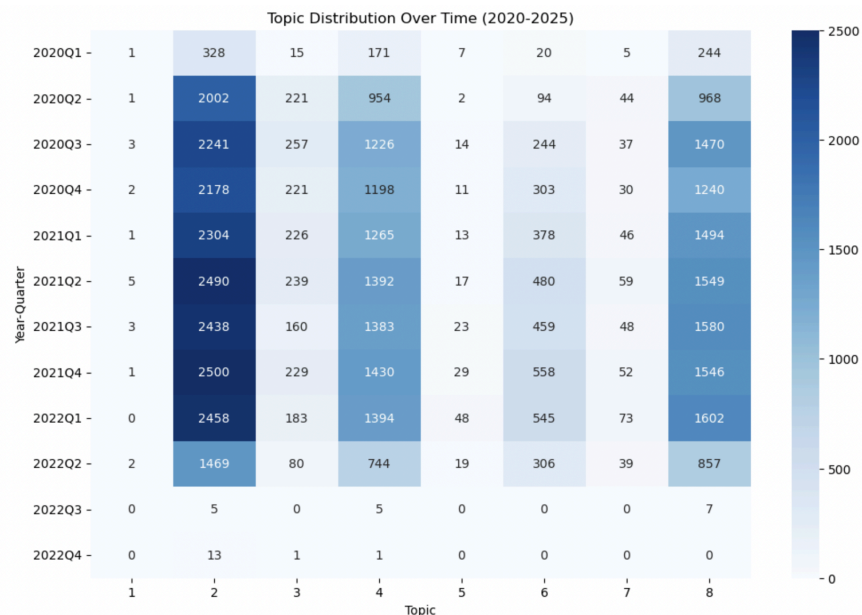
Based on these reduced dimension embeddings, we clustered research documents based on their semantic similarities. We chose MiniBatchKMeans from scikit-learn due to its computational efficiency and scalability. For consistency, the algorithm partitioned the data into 8 clusters (n_clusters=8). The basic assignment of clusters has no interpretive value. KeyBERT was used to automatically generate readable labels for each cluster identified. With the semantic

information encoded in the SciBERT embeddings, keyBERT identifies keywords and keyphrases (including unigrams and bigrams via `keyphrase_ngram_range=(1,2)`) that represent each cluster most accurately. Using the `top_n=3` terms, topic-label descriptions are formed that provide concise yet detailed information, which transform numerical cluster IDs into understandable topics.

Our methodology’s strength lies in its comprehensive approach, which includes detailed data preprocessing, quarterly segmentation, adaptive LDA topic modeling based on coherence scoring, SciBERT, and meticulous parameter tuning. This approach ensured that our analysis accurately captured the evolving nature of COVID-19 research, making our framework a precise and effective tool for tracking changes in the pandemic’s research progress.

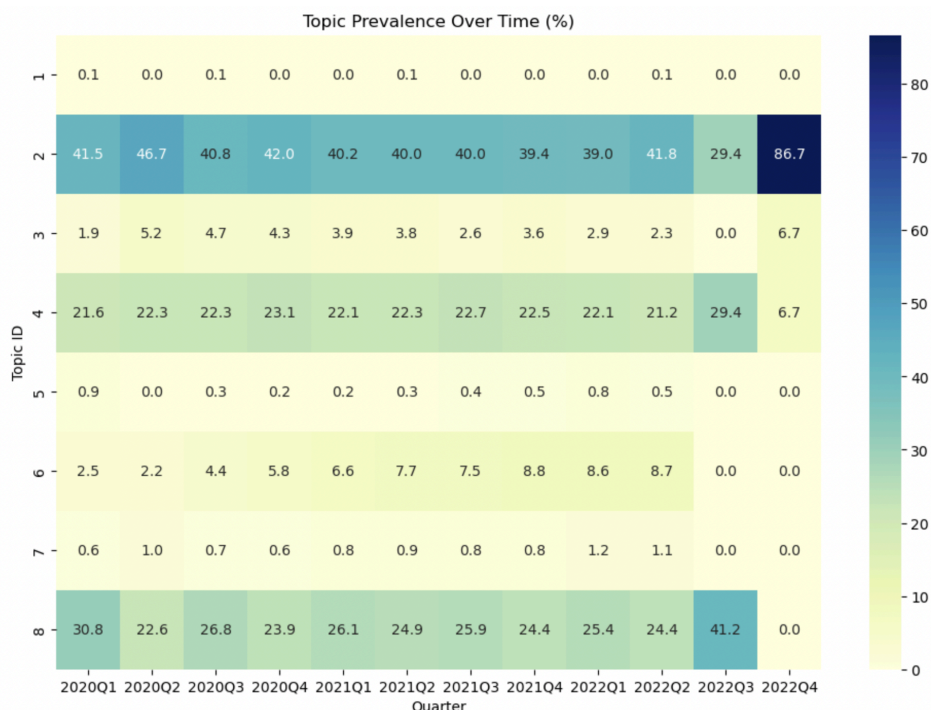
Results and Evaluation

To visualize the evolution of research topics over time, we grouped the dataset by quarter and dominant topic, then used the `groupby()` function to aggregate the data by `year_quarter` and `dominant_topic`. Later, we created a matrix that was normalized into percentages to show the relative prevalence of each topic within each quarter. The “Topic Distribution Over Time” heatmap displays the raw counts of topics per quarter, which provides a clear view of the frequency of each topic over time. The “Topic Prevalence Over Time” heatmap uses the normalized data, which allows us to show the prevalence of each topic over time.



As you can see from the “Topic Distribution Over Time” heatmap, the number of articles related to Topic 2 was 2,500 at its highest in 2021Q4, but declined to just 13 articles by 2022Q4.

This significant change highlights how the frequency of certain topics can fluctuate considerably over time.



As you can see from the “Topic Prevalence Over Time” heatmap, Topic 2 was the most prevalent in 2020Q2 as it represented 46.7% of all articles during that quarter. However by 2022Q3, Topic 8 had become more dominant representing the largest share of 41.2% of articles for that period. This shift in topic prevalence highlights how the focus and importance of different topics evolve over time.

By using SciBERT for embeddings, efficient clustering, and a quarter-based sampling strategy, we were able to visually represent the improved line chart (Figure 2), which depicts the evolution of COVID-19 research topics from 2020 to 2025 based on their article titles. Compared to the earlier version of the line chart (Figure 1), which relied on a predefined/hard-coded list of “magnet words” for clustering, the new and improved method offers a far more data-driven perspective, thus eliminating the biases introduced by manually selecting keywords. Overall, the goal of these charts and graphs is to demonstrate how research focus areas have evolved throughout the pandemic, therefore revealing key patterns into the direction of pandemic-related studies.

Line Chart Analysis & Topic Evolution

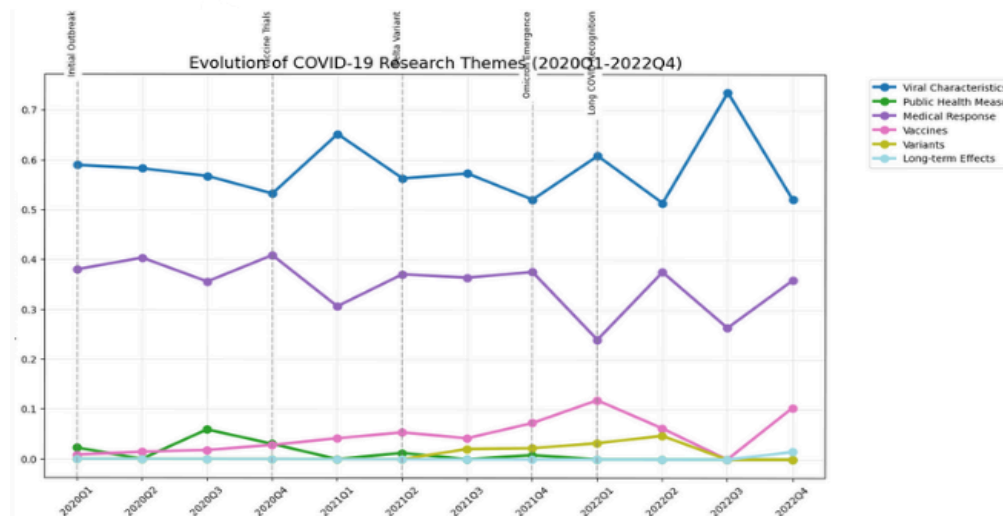


Figure 1: Magnet Word Line Chart

One of the limitations at the forefront of the initial line chart (Figure 1) was its heavy reliance on a hardcoded set of theme-related keywords. This previous implementation manually grouped research topics by clustering them with predefined words such as “vaccine,” “lockdown,” and “variant.” While this approach provided a rough estimate of research trends, it does introduce many biases. For example, the keyword-based grouping failed to capture semantic variations and context which resulted in a rather lop-sided clustering. This leads to fairly noticeable inaccuracies in topic classification. Additionally, when we were experimenting with a pure K-Means algorithm, keywords that did not align neatly with predefined keywords were just thrown in with cluster 1 as seen in Figure 1.1, thus heavily skewing the overall representation of topic prevalence.

Clusters of words:

Cluster 1: based, model, network, problem, analysis, algorithm, system, using, intervention, review, outcome, patient, trial, group, research, clinical, cell, protein, level, model, ace, system, virus, expression, receptor, potential, disease, cov, coronavirus, infection, case, virus, health, respiratory, patient, health, year, high, risk, trauma, using, breathing, infection, virus, human, bat, protein, dog, factor, pcr, patient, day, clinical, severe, co v, infection, case, patient, clinical, treatment, cov, drug, article, cov, group, virus, quarantine, sequence, pressure, may, two, model, case, epidemic number, outbreak, data, virus, china, infection

Cluster 2: study, study, study, study, study, study

Cluster 3: covid, covid, covid, covid

Cluster 4: result, result, result, result, result, result

Cluster 5: sars, sars, sars, sars

Figure 1.1: Sample of Clustering Output using Pure KMeans

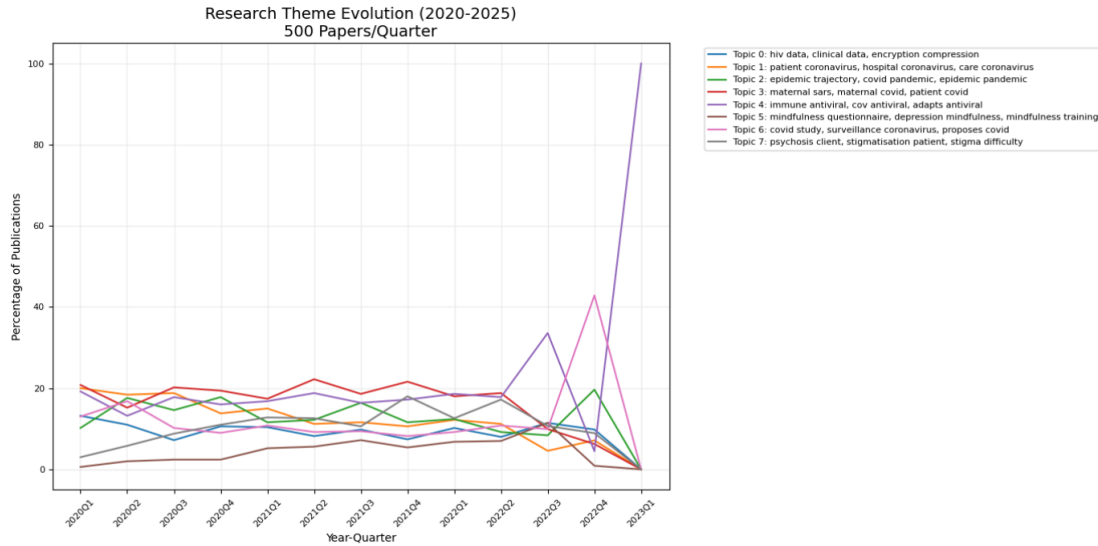


Figure 2: Improved Line Chart using NLP preprocessing, BERT, MiniBatchKMeans, and KEYbert

The improved line chart (Figure 2) overcomes these limitations of clustering by making smart use of SciBERT embeddings and MiniBatchKMeans clustering. Instead of relying on a list of fixed keywords, this new approach clusters research topics based on contextual similarity within a high-dimensional vector space (semantics). This allows for a more flexible and accurate classification of keywords into topics, which means getting closer to presenting the true structure of the research landscape. Additionally, dimensionality reduction via UMAP guarantees computational efficiency without significant loss of information. This improvement has found a good balance between scalability and interpretability.

Going further than methodological improvements, the refined line chart also provides more detailed insights into topic evolution. For example, Topic 3 demonstrates a gradual and consistent rise in early 2020, reflecting researchers' growing interest in drawing parallels with past outbreaks like SARS-COV or examining COVID-19's immediate effects on patients. In contrast, Topic 1 shows a consistent decline over time likely because healthcare systems improved and helped reduce early concerns as more solutions were put in place. These findings demonstrate the value of an unsupervised clustering approach in capturing emergent research trends over time.

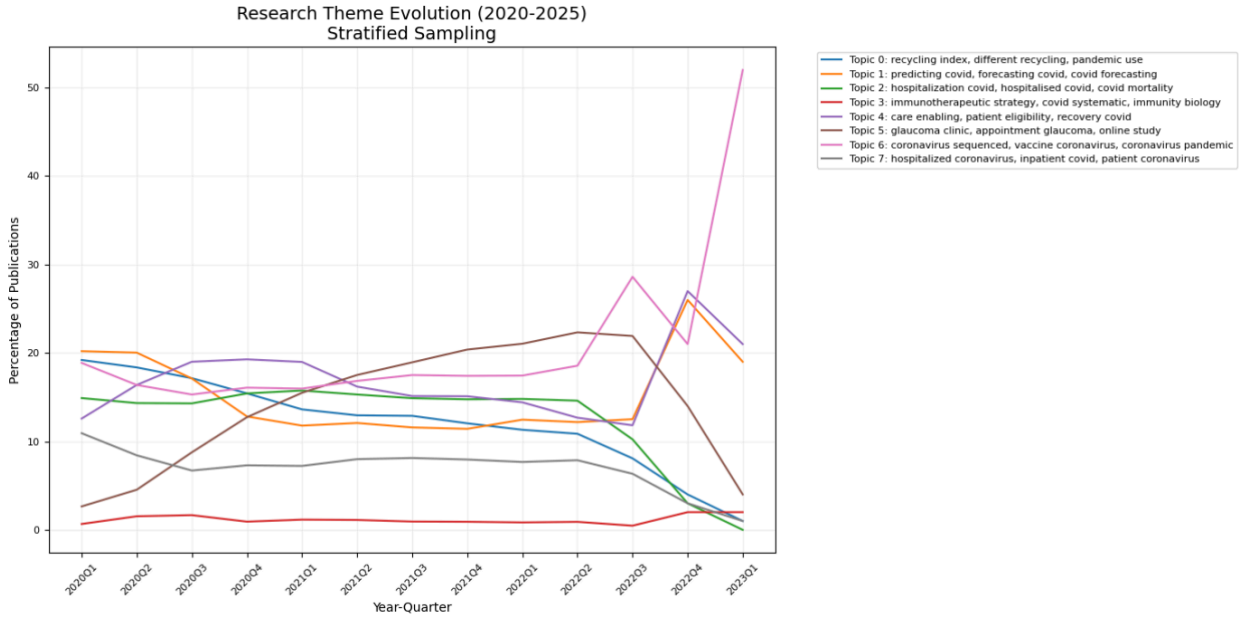


Figure 3: Final iteration of Line Chart using Stratified Sampling

Despite making improvements by using proper data mining techniques like SciBert, and UMAP dimensionality reduction, we still were not satisfied with the fixed 500 article per quarter sampling size. So it was decided to make a final "perfect" iteration of our findings. This final iteration of the line chart uses stratified sampling, where 5% of articles from each quarter are sampled proportionally, instead of manually selecting a fixed number like 500 per quarter as was done in the previous iteration. This new approach offers a more fairly represented and dynamic perspective of the topic distribution over time, especially in a dataset like CORD-19 where the volume of publications rises and falls noticeably from quarter to quarter. By adjusting the sample size proportional to the actual number of articles, stratified sampling avoids over-representing periods with fewer publications and under-representing those with more. This allows for a more precise reflection of topic prevalence and hidden trends.

On the other hand, what happened is that the fixed 500-article sampling from the second iteration introduced a distortion. In other words, quarters with high publication volume had their thematic diversity compressed, while low-volume quarters were artificially inflated in influence. Stratified sampling corrects this imbalance by capturing the true proportion of research themes as they evolve. This solution results in smoother, more trustworthy trends and ensures that sudden spikes or drops in topic popularity are more likely to represent actual shifts in scientific focus rather than artifacts of the sampling method.

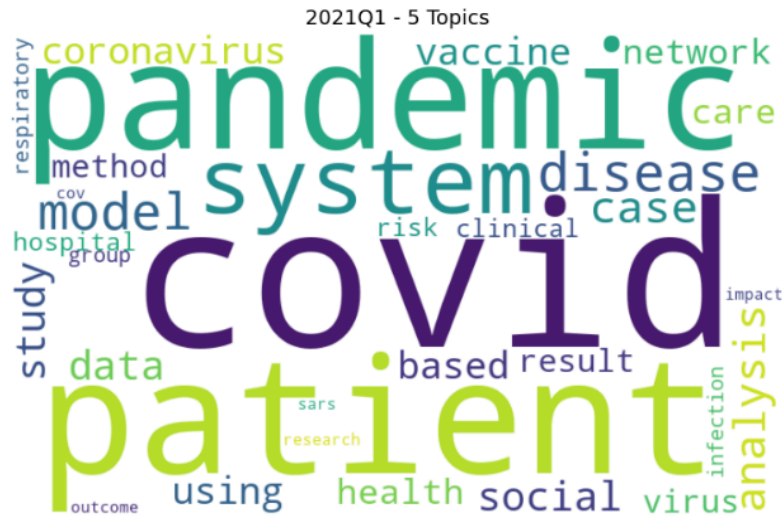
Word Cloud Analysis & Thematic Shifts in Research

Word clouds can serve as an effective yet beautiful method for capturing dominant research themes over different periods. By displaying high-frequency terms in a given timeframe, they provide insight into how the focus of academic discussions evolved throughout 9



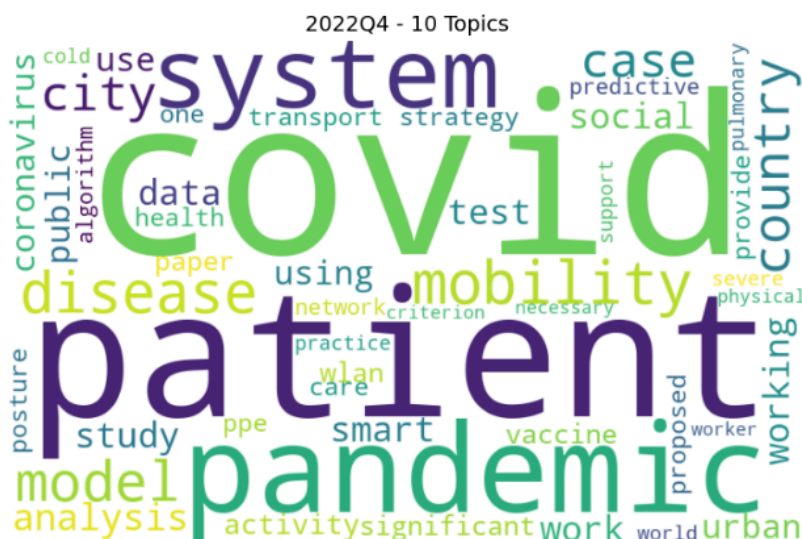
Early Pandemic Research (2020Q1)

The word cloud shows the predominant themes and keywords in COVID-19 research during the first quarter of 2020. The visualization highlights key terms such as "COVID," "system," "health," "case," and "virus," which were central to the scientific research at the start of the pandemic. The size of each word in the cloud represents its frequency across the articles, with larger words representing more frequently occurring terms. This word cloud effectively shows the focus of early COVID-19 research by emphasizing aspects related to "model," "analysis," "data," and "study," which reflected the urgent efforts to understand and model the virus's behavior and impact. The use of this visual tool helps in quickly understanding the range and focus of research topics that were critical during the initial stages of the COVID-19 outbreak.



Collaboration & Vaccination Era (2021Q1)

This word cloud represents the major themes and keywords from COVID-19 research articles published in the first quarter of 2021. Prominent terms such as "pandemic," "COVID," "patient," and "health" dominate the visualization, which reflects the continued focus on both the wider effects of the pandemic and its direct health implications. The presence of words like "vaccine," "clinical," and "care" highlights the shift towards solution-oriented research, including treatment and management of the disease. Smaller yet significant terms such as "model," "data," and "analysis" highlight the scientific community's reliance on data-driven approaches to study the virus's behavior and its effects. This word cloud effectively captures the key research priorities of early 2021 by providing a visual summary of how scientific focus has evolved with the development of new treatments and vaccines.



Long-Term Implications & Post-Pandemic Research (2022Q4)

This word cloud shows the prominent research topics from the fourth quarter of 2022, which highlights the growing diversity in COVID-19 research as the pandemic continues to impact global health and social systems. Central terms such as "COVID," "patient," and "pandemic" remain significant, which indicates ongoing research into direct health impacts and management strategies. The emergence of terms like "mobility," "urban," and "city" reflects a growing focus on the pandemic's effects on urban development and public transportation systems. Additionally, words like "data," "model," and "analysis" emphasize the continued reliance on data-driven methods to understand and predict the behavior of the virus and its broader implications. This word cloud provides an overview of how COVID-19 research has evolved to cover a wider range of themes, which emphasizes the complex relationship between public health measures and societal adaptation.

Overall, word clouds are a great tool that clearly illustrates the evolution in research priorities which lines up with real-time events from crisis containment and epidemiological studies in early 2020, to vaccine campaigns and social policy changes in 2021, and finally to long-term economic and logistical after-effects in 2022. These observations are proof of how the scientific community adapted to the pandemic's unfolding challenges, emphasizing the need for flexible research methodologies that evolve with global crises such as COVID-19.

Future Research Directions and Improvements

Some of our current clustering and visualization techniques can be upgraded to provide more dynamic and accurate insights into topic evolution. For example, the MiniBatchKMeans algorithm used for topic modeling in our code can be upgraded with more sophisticated methods like BERTopic. When added on with dimensionality reduction techniques like UMAP and TF-IDF, it provides improved topic coherence by better capturing the semantic structure of the text. Another option would be an embedding-based approach like EmbedTopic which can directly cluster documents in the embedding space. This reduces the need for predefined cluster counts and better captures subtle topic shifts over time. When paired with interactive visualizations, these sophisticated clustering techniques can make it much easier and more intelligent for researchers to examine the evolution of a topic, which will give a clearer and more comprehensive understanding of scientific research trends.

A promising direction for future research could involve building real-time monitoring tools or dashboards that allow researchers and policymakers to track topic evolution as new publications become available. Such tools could support dynamic filtering, trend projection, and even keyword forecasting based on machine learning models trained on past publication behavior.

Conclusion

When we first started this project, we wanted to explore the evolving topics of scientific research within the CORD-19 dataset, driven by a motivation to help both public health services and the research community with actionable insights. By implementing advanced data mining techniques such as NLTK Preprocessing, LDA Topic Modeling, TF-IDF vectorization, and Heatmap visuals, we successfully mapped the shifting priorities in COVID-19 research. These techniques gave us a more refined and predictive analysis as highlighted by our visualizations which tracked topic evolutions throughout the pandemic. For public health services, our findings highlight critical areas like health and disease research. These findings can help us anticipate future needs and allocate resources effectively for future situations like pandemics. The techniques and structure we applied to the COVID-19 context are also highly transferable to other domains. For example, similar models could be used to track changes in climate-related research, mental health studies, or even technological developments like artificial intelligence. Adapting our topic modeling framework to different datasets would allow us to build research monitoring tools that are both scalable and responsive to shifting academic and societal needs.

Most importantly, this project demonstrates the power of combining data science with real-world relevance. It highlights how the analysis of large-scale research datasets can inform decisions at the intersection of science, policy, and public welfare. Looking ahead, continued refinement of this methodology could help establish a future-ready research environment, where global responses to emerging issues are not just reactive, but strategic and informed by data.

References

Dataset: CORD-19 Dataset

<https://www.kaggle.com/datasets/allen-institute-for-ai/CORD-19-research-challenge/data?select=metadata.csv>